

Distance and Angle Laws of Resolution Geometry

Quotient Statistical Experiments, Kernel-Groupoid Descent, Euclidean Limits, Discriminant Geometry, and No-Go Results

Matěj Rada

MatejRada@email.cz

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Abstract

We close the distance-and-angle rung of Resolution Geometry (RG). The primitive is not delay or latent displacement but a declared statistical experiment together with its equality-of-law quotient and admissibility protocol. On regular finite classical strata, nuisance-efficient Fisher information descends to the observational quotient exactly when it is horizontal and invariant under the full observational kernel groupoid. Its path-length functional gives RG distance and its bilinear polarization gives operational angles. Euclidean geometry is recovered only in a constant isotropic flat chart. The reduction $d = \Theta\gamma M_1$ is the rank-one constant-scale case of $d = \int \Theta\gamma dM$. A first deterministic comparison passed all nine registered checks. A second hostile battery attacks four distinct likelihood families—Bernoulli-logistic, Poisson log-link, categorical softmax, and Gamma shape-scale—and explicit rank, threshold, and orbit-type discriminants. All 28 registered checks pass. Markov garblings contract Fisher information in every applicable family; silent-fibre annihilation, representative invariance, and reparameterization covariance survive all four likelihood structures. At singular loci, the ordinary tangent misses a quadratic quotient direction, whereas a jet-rescaled tangent cone and a $\mathbb{Z}_2$ slice recover the correct one-sided geometry. Hard projectors remain stable off the discriminant but jump by Frobenius norm $\sqrt{2}$ at a threshold exchange. The result strengthens likelihood-family robustness while restricting smooth distance and angle laws to regular strata and replacing them by slice or tangent-cone data at walls.

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1 Scope and dependency discipline

The equality-of-law quotient is forced by the experiment. Fisher geometry, nuisance convention, thresholds, transport, and scale are later modules. The paper does not derive a metric from the kernel pair alone. It characterizes the metric law after the regular classical statistical module is declared and tests the precise gates under which that law descends, reduces, and fails.

2 Declared quotient experiment

Definition 2.1 (Complete experiment). A complete statistical experiment is $\mathcal{E} = (\Theta, (\Omega, \mathcal{A}), \{P_\theta\}_{\theta \in \Theta})$. Define

$$\theta \sim_{\mathcal{E}} \theta' \iff P_\theta = P_{\theta'}, \quad Q_{\mathcal{E}} = \Theta / \sim_{\mathcal{E}},$$

with quotient map $\pi : \Theta \rightarrow Q_{\mathcal{E}}$ and kernel groupoid

$$\mathcal{K}_{\mathcal{E}} = \Theta \times_{Q_{\mathcal{E}}} \Theta \rightrightarrows \Theta.$$

Definition 2.2 (Regular score module). On a dominated regular stratum, let $S_u = \partial_u \log p(Y \mid \theta)$. For target coordinates q and nuisance coordinates ν , partition

$$I = \begin{pmatrix} I_{qq} & I_{q\nu} \\ I_{\nu q} & I_{\nu\nu} \end{pmatrix}, \quad g_{\text{eff}} = I_{qq} - I_{q\nu} I_{\nu\nu}^+ I_{\nu q}.$$

3 Kernel-groupoid descent

Theorem 3.1 (Kernel-groupoid metric descent). *On a regular quotient stratum, a symmetric positive-semidefinite tensor g on Θ defines a unique tensor $\bar{g}$ on $Q_{\mathcal{E}}$ if and only if:*

1. $g(v, \cdot) = 0$ for every $v \in \ker D\pi$;
2. g is invariant under arrows of $\mathcal{K}_{\mathcal{E}}$ under the declared horizontal identifications.

Then $g = \pi^ \bar{g}$ on the regular stratum.*

Proof. A pullback annihilates vertical vectors and is representative-independent. Conversely, horizontality makes $g_\theta(u, v)$ depend only on $D\pi_\theta u, D\pi_\theta v$, and groupoid invariance removes dependence on the representative. Surjectivity of π gives uniqueness. $\square$

Corollary 3.2 (Declared groupoid criterion). *A sound declared groupoid $A \hookrightarrow \mathcal{K}_{\mathcal{E}}$ suffices for metric descent only when it is gauge-complete, $A \cdot \theta = [\theta]_{\mathcal{E}}$.*

No-Go Theorem 3.3 (Kernel pair does not select a metric). The quotient and kernel groupoid determine observational classes and universal factorization, but do not determine g_{eff} , normalization, nuisance convention, or physical scale.

4 Distance and angle laws

Definition 4.1 (RG length and distance). For absolutely continuous $\Gamma : [0, 1] \rightarrow Q_{\mathcal{E}}$,

$$L_{\text{RG}}[\Gamma] = \int_0^1 \sqrt{\bar{g}_{\Gamma(s)}(\dot{\Gamma}, \dot{\Gamma})} ds, \quad d_{\text{RG}}(q_0, q_1) = \inf_{\Gamma: q_0 \rightarrow q_1} L_{\text{RG}}[\Gamma].$$

Theorem 4.2 (Observable distance law). *If g_{eff} is kernel-groupoid basic and positive definite on a connected regular quotient stratum, d_{RG} is the intrinsic Riemannian distance of the nuisance-efficient experiment and is invariant under admissible reparameterization and natural experiment isomorphism.*

Definition 4.3 (Operational angle). For nonzero $u, v \in T_q Q_{\mathcal{E}}$,

$$\cos \angle_q(u, v) = \frac{\bar{g}_q(u, v)}{\sqrt{\bar{g}_q(u, u)\bar{g}_q(v, v)}}.$$

No-Go Theorem 4.4 (Rank-one angle no-go). A rank-one experiment defines length and order along its sole resolved direction but no intrinsic angle between two independent quotient directions.

5 The reduction $d = \Theta\gamma M_1$

Theorem 5.1 (Rank-one reduction). *Suppose a regular quotient channel admits coordinate M and*

$$\bar{g} = (\Theta(M)\gamma(M))^2 dM^2.$$

Then

$$d_{\text{RG}}(M_0, M_1) = \left| \int_{M_0}^{M_1} \Theta(M)\gamma(M) dM \right|.$$

If $\Theta\gamma$ is constant and $M_0 = 0$, then

$$\boxed{d = \Theta\gamma M_1}.$$

Corollary 5.2 (Heterogeneous correction). *When $\Theta\gamma$ varies, a global product is generally biased. The comparison control produced a 5.10% discrepancy.*

Remark 5.3 (Moment naming). Here M_1 denotes accumulated resolved measure. If it denotes a temporal first moment, a time weight must appear explicitly.

6 Euclidean limit and failure law

Theorem 6.1 (Euclidean recovery). *On a simply connected chart, if $\bar{g} = \lambda^2 I$ with constant $\lambda > 0$, then*

$$d_{\text{RG}}(q_0, q_1) = \lambda \|q_1 - q_0\|_2,$$

geodesics are straight segments and RG angles equal Euclidean angles.

Corollary 6.2 (Calibrated Euclidean identification). *After an independent scale calibration setting $\lambda = 1$, RG distance equals Euclidean distance on that chart.*

No-Go Theorem 6.3 (Euclidean-identification no-go). Flatness alone does not imply physical Euclidean distance. Anisotropy, nonconstant conformal factor, heterogeneous information density, unresolved nuisance, rank change, nonbasicness, or missing scale calibration breaks the identification.

7 Ignorance and experiment comparison

Theorem 7.1 (Garbling contraction). *If $\mathcal{F}$ is obtained from $\mathcal{E}$ by a parameter-independent Markov kernel, then*

$$g_{\mathcal{F}} \preceq g_{\mathcal{E}}.$$

Corresponding path lengths and induced infimum distance cannot increase.

Corollary 7.2 (Ignorance collapse). *If garbling removes a score direction, that direction enters the radical of the processed Fisher form. Distinct fine-experiment states may have zero distance in the coarse quotient.*

No-Go Theorem 7.3 (Equal quotient is not equal experiment). Equality of identifiability kernel pairs does not imply Blackwell equivalence. The quotient alone cannot determine quantitative distance loss under processing.

8 Likelihood-family universality attack

The previous comparison used Gaussian means. We therefore attacked four different regular likelihood families.

Definition 8.1 (Multi-family class). The class comprises independent Bernoulli observations with logistic parameters, independent Poisson counts with a log link, categorical softmax laws, and a Gamma shape–scale family in logarithmic coordinates. In each family, a coordinate h was appended but omitted from the outcome law, creating an exact synthetic observation fibre.

Proposition 8.2 (Family-independent basicness). *For all four families the augmented Fisher tensor was*

$$\tilde{g} = \begin{pmatrix} g_{\text{family}} & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus $\tilde{g}(\partial_h, \cdot) = 0$ and changing h leaves the horizontal tensor unchanged. The recorded vertical norm and representative discrepancy were zero in every family.

Proposition 8.3 (Cross-family processing and covariance). *The tested garblings obeyed $g_{\text{garbled}} \preceq g_{\text{original}}$. The largest generalized Fisher ratios were 0.312049 for binary-symmetric Bernoulli garbling, exactly 0.37 for Poisson thinning with retention 0.37, and 1 for the categorical merge. Under $\theta' = S\theta$, every family also obeyed*

$$g' = S^{-\top} g S^{-1}, \quad u^{\top} g u = (Su)^{\top} g' (Su)$$

to numerical precision.

Corollary 8.4 (Likelihood-family robustness). *Within the attacked regular classical domain, horizontality, representative invariance, Markov contraction, and parameter covariance are properties of the statistical-experiment construction rather than artefacts of a Gaussian mean likelihood.*

Non-claim 8.5. Four finite regular families do not prove the corresponding statement for every non-dominated, singular, infinite-dimensional, quantum, or process-valued experiment.

9 Discriminant stress tests

9.1 Structural and protocol rank walls

For

$$\mu_\varepsilon(x, y) = (x, \varepsilon y), \quad g_\varepsilon = \text{diag}(1, \varepsilon^2),$$

the exact rank is two for $\varepsilon \neq 0$ and one at $\varepsilon = 0$. The battery recovered $\lambda_{\min} = \varepsilon^2$ with log-log slope 2.

Proposition 9.1 (Structural versus finite-resolution wall). *The structural wall is $\varepsilon = 0$. A declared tolerance $\tau > 0$ creates a protocol wall at $\varepsilon^2 \leq \tau$. A resolution-dependent rank loss can therefore precede the exact wall.*

9.2 Reflection slice and tangent cone

Let $\mathbb{Z}_2$ act by $(u, v) \mapsto (u, -v)$ and define

$$\pi(u, v) = (u, r), \quad r = v^2.$$

The quotient is locally $r \geq 0$.

Proposition 9.2 (Slice construction). *The half-slice $v \geq 0$ intersects every nonzero orbit once and preserves $r = v^2$. On $r > 0$,*

$$dv^2 = \frac{dr^2}{4r}.$$

The coefficient diverges at $r = 0$, so this regular-coordinate metric does not extend smoothly through the orbit-type wall.

Theorem 9.3 (Jet-rescaled tangent-cone recovery). *At the wall,*

$$D\pi_{(0,0)} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

has rank one. For arcs $(u(h), v(h)) = (ha, hb)$,

$$\left(\frac{\Delta u}{h}, \frac{\Delta r}{h^2} \right) \rightarrow (a, b^2).$$

Hence the jet-resolved tangent cone is

$$C_{(0,0)}Q = \mathbb{R} \times \mathbb{R}_{\geq 0}.$$

No half-space violations occurred in 5000 random secants, and 2000 sampled cone directions were constructively recovered to machine precision.

No-Go Theorem 9.4 (Ordinary-tangent continuation no-go). When the first nonzero transverse quotient invariant is quadratic, $D\pi$ alone misses a genuine one-sided observable direction. A slice, tangent cone, or higher-jet construction is required.

9.3 Hard threshold crossing

Let

$$L(t) = \text{diag}(1+t, 1-t, 0.2), \quad \tau = 1, \quad P_\tau(t) = \mathbf{1}_{(1,\infty)}(L(t)).$$

For $t > 0$ the first direction is selected; for $t < 0$ the second is selected; at $t = 0$ strict thresholding selects neither threshold eigenvector.

Proposition 9.5 (Projector wall and off-wall stability). *The one-sided projectors obey*

$$\|P_\tau(0^+) - P_\tau(0^-)\|_F = \sqrt{2}.$$

Thus equal one-sided rank does not prevent a discontinuous subspace exchange. Conversely, 1000 perturbations bounded by one quarter of a positive spectral margin produced no projector change off the wall.

Corollary 9.6 (Strengthened stratum-locality). *Smooth distance and angle tensors apply only on constant-rank, fixed-orbit-type, threshold-separated strata. At rank walls use stratified length completion; at isotropy walls use slices and tangent cones; at hard threshold exchange retain one-sided sectors unless an additional gluing law is supplied.*

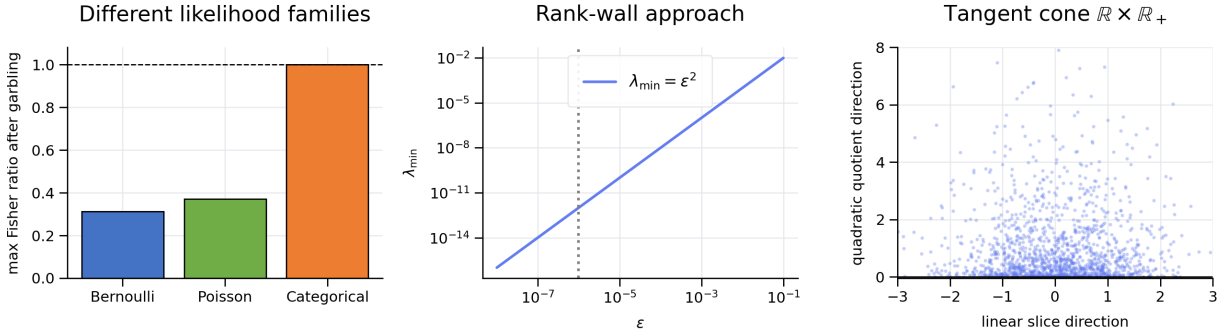


Figure 1: Left: Fisher contraction across different likelihood families. Centre: quadratic closing of the smallest information eigenvalue. Right: jet-rescaled secants fill $\mathbb{R} \times \mathbb{R}_{\geq 0}$.

10 Deterministic stress-test ledger

The original Euclidean comparison passed nine of nine checks. The second battery used deterministic seed 20260927 and passed all 28 registered checks.

Stress test	Recorded value	Gate	Result
Bernoulli garbling ratio	0.312049	≤ 1	pass
Poisson thinning ratio	0.370000	$= \alpha$	pass
Categorical merge ratio	1.000000	≤ 1	pass
Rank-wall log-log slope	2.000000	quadratic	pass
Tangent-cone violations	0	none	pass
Cone reconstruction error	8.88×10^{-16}	$< 10^{-12}$	pass
Threshold projector jump	$\sqrt{2}$	exact	pass
Off-wall projector failures	0/1000	none	pass

Table 1: Selected outcomes from the 28-check multi-family discriminant battery.

11 Final theorem ladder insertion

Theorem 11.1 (Distance-angle closure theorem). *Let $(\mathcal{E}, \Gamma)$ be a regular finite classical RG experiment. Assume equality-of-law quotienting, a declared nuisance convention, the Fisher metric module, kernel-groupoid basicness, positive definiteness on the selected stratum, natural chart transport, and independent scale normalization when physical units are claimed. Then*

$$d_{\text{RG}} = \inf_{\Gamma} \int \sqrt{g_{\text{eff}}(\dot{\Gamma}, \dot{\Gamma})}, \quad \cos \alpha = \frac{g_{\text{eff}}(u, v)}{\|u\|_{g_{\text{eff}}} \|v\|_{g_{\text{eff}}}}.$$

The law is monotone under garbling, invariant under experiment isomorphism, and collapses on persistent observational null directions. Its rank-one constant-scale branch is $d = \Theta\gamma M_1$ and its constant isotropic calibrated branch is Euclidean geometry. These gates survived four distinct regular likelihood families. At rank, threshold, or orbit-type discriminants, the smooth law is replaced by stratumwise limits together with declared slice, tangent-cone, or one-sided projector data; no single smooth continuation is forced.

12 Non-claims and falsification criteria

Non-claim 12.1. The kernel pair does not derive Fisher geometry, nuisance conventions, or physical units.

Non-claim 12.2. Statistical distance is experiment-relative and is not automatically latent spatial distance.

Non-claim 12.3. A Euclidean calibration on one chart does not globalize across rank walls or heterogeneous media without gluing and scale-integrability tests.

Non-claim 12.4. The battery does not validate a particular underwater, seismic, optical, or wall experiment. Physical identification requires empirical likelihoods, covariance, nuisance budgets, held-out prediction, and scale anchors.

The characterization is falsified if distance varies inside an equality-of-law fibre, increases under genuine garbling, changes under correctly transported experiment isomorphism, assigns a nondegenerate angle to a rank-one stratum, uses $\Theta\gamma M_1$ with variable scale, or continues an ordinary tangent smoothly through a wall where the quotient requires a cone or slice.

13 Conclusion

RG distance is the intrinsic length of the nuisance-efficient distinguishability tensor after descent through the full observational kernel groupoid. Angles are normalized cross-information. Ignorance enlarges fibres and contracts or annihilates lengths. Euclidean geometry is recovered as a constant isotropic calibrated limit. The reduction $d = \Theta\gamma M_1$ is legitimate only on a rank-one, groupoid-basic, constant-scale regular channel. The multi-family attack shows that the gates are not confined to Gaussian means; the discriminant attack shows that the regular law must stop honestly at walls and hand over to stratified slice and tangent-cone geometry.

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